QP Code: 26900347

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours)
Regular Examination – October 2026
Fifth Semester
MG5DSCMAT301 – Limits and Convergence

Scheme of Valuation

General Instructions for Valuation

- Part A: 5 questions to be answered. The total marks for Part A shall not exceed 20 marks.
- Part B: 5 questions to be answered. The total marks for Part B shall not exceed 30 marks.
- Part C: 3 questions to be answered. The total marks for Part C shall not exceed 30 marks.
- Award full marks for a complete and correct alternate answer. Partial marks should be awarded for partially correct alternate answers based on the evaluator's discretion.

Part A: Short Answer Type Questions

Answer any 5 questions. (The total marks for Part A shall not exceed 20 marks.)

1. Find

$$\lim_{n\to\infty}\left(\frac{2n+1}{n}\right).$$

[A]/[CO1]

Solution:

$$\lim_{n\to\infty}\frac{2n+1}{n}=\lim_{n\to\infty}\left(2+\frac{1}{n}\right)=2.$$

Note: Full marks can be given for other answers that use the definition of the limit of a sequence. (3 marks)

2. Write the tail of the sequence $\left(\frac{1}{n}\right)$ beginning with x_5. [U]/[CO1]

Solution: The tail beginning with x_5 is

$$\left(\frac{1}{5},\frac{1}{6},\frac{1}{7},\frac{1}{8},\dots\right).$$

(2 marks)

3. Why is the boundedness condition essential in the Bolzano–Weierstrass Theorem? [U]/[CO3]

Solution: The Bolzano–Weierstrass Theorem states that every bounded sequence of real numbers has a convergent subsequence. Boundedness is essential because an unbounded sequence need not have a convergent subsequence. (2 marks)

4. State the theorem on the proper divergence of monotone sequences. What can be concluded if a monotone sequence is unbounded above or unbounded below? [U]/[CO5]

Solution: A monotone sequence that is unbounded above diverges to $+\infty$, while a monotone sequence that is unbounded below diverges to $-\infty$. (2 marks)

5. Prove that the ratio test fails to check the convergence of the p-series. [An]/[CO6]

Solution: Consider the p-series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}.$$

Here, $x_n = \frac{1}{n^p}$. By the ratio test,

$$\begin{aligned} L &= \lim_{n\to\infty} \left| \frac{x_{n+1}}{x_n} \right| \\ &= \lim_{n\to\infty} \frac{n^p}{(n+1)^p} \\ &= \lim_{n\to\infty} \left(\frac{n}{n+1} \right)^p \\ &= 1. \end{aligned}$$

Since $L = 1$, the ratio test is inconclusive and cannot determine the convergence or divergence of the p-series.

(3 marks)

6. Discuss the conditions needed for Abel's Test. [U]/[CO7]

Solution: Abel's Test states that the series $\sum_{n=1}^{\infty} x_n y_n$ converges if:

(a) The sequence (x_n) is a convergent monotone sequence.

(b) The series $\sum_{n=1}^{\infty} y_n$ is convergent.

(3 marks)

7. Explain the meaning of $\lim_{x\to-\infty} f(x) = L$. [U]/[CO8]

Solution: Definition

(2 marks)

8. What is the limit of the function $f(x) = \frac{\sin x}{x}$ as $x \to 0$? [U]/[CO8]

Solution: The limit is

$$\lim_{x\to 0} \frac{\sin x}{x} = 1.$$

(2 marks)

Part B: Short Essay Type Questions

Answer any 5 questions. (The total marks for Part B shall not exceed 30 marks.)

9. Prove that every convergent sequence of real numbers is bounded. [A]/[CO1]

Solution:

Suppose that $\lim_{n\to\infty} x_n = x$ and let $\varepsilon := 1$. (1 Mark)

Then there exists a natural number $K = K(1)$ such that

$$|x_n - x| < 1 \quad \text{for all } n \geq K.$$

(2 Marks)

If we apply the Triangle Inequality with $n \geq K$, we obtain

$$\begin{aligned} |x_n| &= |x_n - x + x| \\ &\leq |x_n - x| + |x| \\ &< 1 + |x|. \end{aligned}$$

(3 Marks)

If we set

$$M := \sup\{|x_1|, |x_2|, \ldots, |x_{K-1}|, 1 + |x|\},$$

(1 Mark)

then it follows that

$$|x_n| \leq M \quad \text{for all } n \in \mathbb{N}.$$

Hence, (x_n) is bounded. (1 Mark)

Total = 8 Marks

10. Prove that if (x_n) is a bounded increasing sequence, then

$$\lim(x_n) = \sup\{x_n : n \in \mathbb{N}\}.$$

[A]/[CO2]

Solution:

Let (x_n) be a bounded, increasing sequence. Since (x_n) is bounded, there exists $M \in \mathbb{R}$ such that

$$x_n \leq M \quad \text{for all } n \in \mathbb{N}.$$

(1 Mark)

By the completeness property of $\mathbb{R}$, the supremum

$$x^* = \sup\{x_n : n \in \mathbb{N}\}$$

exists in $\mathbb{R}$. (2 Marks)

Let $\varepsilon > 0$ be given. Then $x^* - \varepsilon$ is not an upper bound of the set $\{x_n : n \in \mathbb{N}\}$.

Hence, there exists $K \in \mathbb{N}$ such that

$$x^* - \varepsilon < x_K.$$

(2 Marks)

Since (x_n) is increasing, for every $n \geq K$,

$$x_K \leq x_n.$$

Also, since x^* is the supremum,

$$x_n \leq x^*.$$

Combining these inequalities, we obtain

$$x^* - \varepsilon < x_K \leq x_n \leq x^* < x^* + \varepsilon$$

for all $n \geq K$. (2 Marks)

Therefore,

$$|x_n - x^*| < \varepsilon \quad \text{for all } n \geq K.$$

Since $\varepsilon > 0$ is arbitrary, we conclude that

$$\lim_{n\to\infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

(1 Mark)

Total = 8 Marks

11. Let (x_n) and (y_n) be two sequences of positive real numbers and suppose that for some $L \in R, L \geq 0$, $lim(\frac{x_n}{y_n}) = L$.Using the definition of a properly divergent sequence, show that if $\lim(y_n) = +\infty$, then $\lim(x_n) = +\infty$. [A]/[CO5]

Solution:

$\lim_{n\to\infty}\left(\frac{x_n}{y_n}\right) = L$ holds, there exists $K \in \mathbb{N}$ such that

$$\frac{1}{2}L < \frac{x_n}{y_n} < \frac{3}{2}L \quad \text{for all } n \geq K.$$

(3 Marks)

Hence we have

$$\left(\frac{1}{2}L\right) y_n < x_n < \left(\frac{3}{2}L\right) y_n \quad \text{for all } n \geq K.$$

(3 Marks)

12(a). Define a Cauchy sequence.
(b) If $x_1 = 1, \quad x_2 = 2, \quad x_n = \frac{1}{2}(x_{n-2} - x_{n-1})$ for $n > 2$, prove that $|x_n - x_{n+1}| = \frac{1}{2^{n-1}}$ for $n \in R$. [A]/[CO4]

Solution:

(a) A sequence $X = (x_n)$ of real numbers is said to be a Cauchy sequence if for every $\varepsilon > 0$ there exists a natural number $H(\varepsilon)$ such that for all natural numbers $n, m \geq H(\varepsilon)$, the terms x_n, x_m satisfy $|x_n - x_m| < \varepsilon$. (3 Marks)
(b) Any reasonable attempt (3 Marks)

13. Prove that absolute convergence of a series implies convergence, but not conversely. [An]/[CO6]

Solution:

Since $\sum_{n=1}^{\infty} |x_n|$ is convergent, the Cauchy Criterion implies that, given $\varepsilon > 0$, there exists $M(\varepsilon) \in \mathbb{N}$ such that if $m > n \geq M(\varepsilon)$, then

$$|x_{n+1}| + |x_{n+2}| + \cdots + |x_m| < \varepsilon.$$

(3 Marks)

By Triangle Inequality,

$$\begin{aligned} |s_m - s_n| &= |x_{n+1} + x_{n+2} + \cdots + x_m| \\ &\leq |x_{n+1}| + |x_{n+2}| + \cdots + |x_m| \\ &< \varepsilon. \end{aligned}$$

(2 Marks)

Since $\varepsilon > 0$ is arbitrary, the Cauchy Criterion implies that $\sum_{n=1}^{\infty} x_n$ converges.

(1 Mark)

To show that the converse is false, consider

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}.$$

This alternating harmonic series converges by the alternating series test. However, its absolute series is

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^{n+1}}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n},$$

which diverges. (2 Mark)

Total = 8 Marks

14. Use Raabe's test to determine whether the series

$$\sum_{n=1}^{\infty} \frac{1}{n(n+3)}$$

converges or diverges. [U]/[CO6]

Solution:

Let

$$x_n = \frac{1}{n(n+3)}.$$

(1 Mark)

By Raabe's test, calculate

$$R = \lim_{n \to \infty} n \left(\frac{x_n}{x_{n+1}} - 1 \right).$$

(2 Marks)

Now,

$$x_{n+1} = \frac{1}{(n+1)(n+4)}.$$

(1 Mark)

Hence,

$$\frac{x_n}{x_{n+1}} = \frac{(n+1)(n+4)}{n(n+3)}.$$

(2 Marks)

Therefore,

$$\begin{aligned} R &= \lim_{n\to\infty} n\left(\frac{(n+1)(n+4)}{n(n+3)} - 1\right) \\ &= \lim_{n\to\infty} \frac{2n+4}{n+3} \\ &= 2. \end{aligned}$$

Since $R = 2 > 1$, Raabe's test gives convergence.

Thus, $\sum_{n=1}^{\infty} \frac{1}{n(n+3)}$ converges. (2 Marks)

Total = 8 Marks

15. Prove the squeeze theorem for functions. [K]/[CO8]

Solution:

Let $A \subseteq \mathbb{R}$, let $f, g, h : A \to \mathbb{R}$, and let $c \in \mathbb{R}$ be a cluster point of A. If

$$f(x) \leq g(x) \leq h(x) \quad \text{for all } x \in A,\ x \neq c,$$

and if

$$\lim_{x\to c} f = L = \lim_{x\to c} h,$$

then

$$\lim_{x\to c} g = L.$$

(4 Marks)

Proof (2 Marks)

16. Explain the limit theorem for quotient of two functions and explain why it cannot be directly use to find the limit of $\frac{x^2-4}{3x-6}$ at $x = 2$. Also find the limit. [U]/[CO8]

Solution:

The quotient limit theorem states that if

$$\lim_{x\to a} f(x) = L, \qquad \lim_{x\to a} g(x) = M, \qquad M \neq 0,$$

then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

(4 Marks)

Consider

$$\lim_{x\to 2} \frac{x^2-4}{3x-6}.$$

Direct substitution gives

$$\frac{2^2-4}{3(2)-6} = \frac{0}{0}.$$

The quotient limit theorem cannot be applied directly because the limit of the denominator is zero.

Factor the numerator and denominator:

$$\frac{x^2-4}{3x-6} = \frac{(x-2)(x+2)}{3(x-2)}.$$

For $x \neq 2$, this simplifies to

$$\frac{x+2}{3}.$$

Therefore,

$$\begin{aligned}\lim_{x\to 2} \frac{x^2-4}{3x-6} &= \lim_{x\to 2} \frac{x+2}{3} \\ &= \frac{2+2}{3} \\ &= \frac{4}{3}.\end{aligned}$$

(4 Marks)

Total = 8 Marks

Part C: Essay Type Questions

Answer any 3 questions. (The total marks for Part C shall not exceed 30 marks.)

17. Use the definition of the limit of a sequence to prove the following limits.

(a)

$$\lim\,(\sqrt{n+1} - \sqrt{n}) = 0.$$

(b)

$$\lim\,(\frac{3n+1}{n+1}) = 3.$$

[A]/[CO1]

Solution: (a)Let $\varepsilon > 0$ be given. We have

$$|\sqrt{n+1} - \sqrt{n}| = \frac{|(\sqrt{n+1} - \sqrt{n})(\sqrt{n+1} + \sqrt{n})|}{\sqrt{n+1} + \sqrt{n}}$$
$$= \frac{1}{\sqrt{n+1} + \sqrt{n}}$$

$$< \frac{1}{\sqrt{n}}.$$

(6 Marks)

(b) Let $\varepsilon > 0$ be given. We have

$$\left|\frac{3n+1}{n+1} - 3\right| = \left|\frac{3n+1-3n-3}{n+1}\right|$$
$$= \frac{2}{n+1}.$$

(4 Marks)

Choose $N \in \mathbb{N}$ such that

$$N > \frac{2}{\varepsilon}.$$

Then, for every $n \geq N$,

$$\left|\frac{3n+1}{n+1} - 3\right| = \frac{2}{n+1} < \varepsilon.$$

Therefore,

$$\lim_{n\to\infty} \frac{3n+1}{n+1} = 3.$$

(2 Marks)

Total = 12 Marks

18. (a) Prove that every Cauchy sequence of real numbers is bounded. Is the converse true?
(b) State and prove the Cauchy criterion for convergence of a sequence.
[An]/[CO4]

Solution: (a)Let $X = (x_n)$ be a Cauchy sequence and let $\varepsilon = 1$. (2 Marks)
There exists $H \in \mathbb{N}$ such that, for all $n \geq H$,

$$|x_n - x_H| < 1.$$

(2 Marks)

By the triangle inequality,

$$|x_n| \leq |x_n - x_H| + |x_H| < 1 + |x_H|.$$

(1 Mark)

Let

$$M = \sup\{|x_1|, |x_2|, \ldots, |x_{H-1}|, |x_H| + 1\}.$$

Then

$$|x_n| \leq M \quad \text{for all } n \in \mathbb{N}.$$

Hence, (x_n) is bounded. (1 Mark)
The converse is false. (2 Mark)

(b) Statement: A sequence of real numbers is convergent if and only if it is a Cauchy sequence. (2 Marks)
Necessity: Suppose $x_n \to x$. For every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|x_n - x| < \frac{\varepsilon}{2}, \quad n \geq N.$$

Hence, for $m, n \geq N$,

$$|x_n - x_m| \leq |x_n - x| + |x_m - x| < \varepsilon.$$

Thus, (x_n) is a Cauchy sequence. (3 Marks)
Sufficiency: Suppose (x_n) is a Cauchy sequence. It is bounded, so by the Bolzano–Weierstrass theorem, it has a convergent subsequence (x_{n_k}) such that $x_{n_k} \to x$. Since (x_n) is Cauchy, the entire sequence converges to x. (2 Marks)

Total = 15 Marks

19. Let $X = (x_n)$ be a sequence of real numbers. Then prove that the following statements are equivalent. (i) The sequence $X = (x_n)$ does not converge to $x \in \mathbb{R}$. (ii) There exists an $\varepsilon_0 > 0$ such that for any $k \in \mathbb{N}$, there exists $n_k \in N$ such that $n_k \geq k$ and

$$|x_{n_k} - x| \geq \varepsilon_0.$$

(iii) There exists an $\varepsilon_0 > 0$ and a subsequence $X' = (x_{n_k})$ of X such that

$$|x_{n_k} - x| \geq \varepsilon_0 \quad \text{for all } k \in \mathbb{N}.$$

[An]/[CO3]
Solution: (i) implies (ii) (4 Marks)
(ii) implies (iii) (3 Marks)
(iii) implies (i) (3 Marks)

20. Discuss the convergence and divergence of the geometric series. [A]/[CO5]
Solution: A geometric series is of the form

$$a + ar + ar^2 + \cdots = \sum_{n=0}^{\infty} ar^n,$$

where a is the first term and r is the common ratio. (2 Marks)

The nth partial sum is $S_n = a + ar + \cdots + ar^n$. (2 Marks)

$$S_n = \frac{a(1 - r^{n+1})}{1 - r}.$$

(2 Marks)

Case 1: $|r| < 1$. The geometric series converges and

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1 - r}.$$

(2 Marks)

Case 2: $|r| > 1$. The series diverges. (2 Marks)

Case 3: $r = 1$. The series diverges. (1 Mark)

Case 4: $r = -1$. The series diverges. (1 Mark)

Total = 12 Marks

21. (a) Describe the right-hand and left-hand limits of $f(x) = \text{sgn}(x)$ at $x = 0$. and state whether they are distinct

(b) Describe the behaviour of the one-sided limits of $g(x) = e^{1/x}$ at $x = 0$. [U]/[CO8]

Solution: (a) The sign function is defined by

$$\text{sgn}(x) = \begin{cases} -1, & x < 0, \\ 0, & x = 0, \\ 1, & x > 0. \end{cases}$$

(5 Marks)

For $x > 0$, $f(x) = 1$. Therefore,

$$\lim_{x \to 0^+} f(x) = 1.$$

(1 Mark)

For $x < 0$, $f(x) = -1$. Therefore,

$$\lim_{x \to 0^-} f(x) = -1.$$

(1 Mark)

(b) As $x \to 0^+$,

$$\lim_{x \to 0^+} e^{1/x} = +\infty.$$

(4 Marks)

As $x \to 0^-$,

$$\lim_{x \to 0^-} e^{1/x} = 0.$$

(4 Marks)

Total = 15 Marks

22. Let $A \subseteq \mathbb{R}$, $f : A \to \mathbb{R}$, and let c be a cluster point of A. Then prove that the following statements are equivalent. (i)$\lim_{x\to c} f(x) = L$. (ii) For every sequence (x_n) in A that converges to c such that $x_n \neq c$ for all $n \in \mathbb{N}$, the sequence $(f(x_n))$ converges to L.

Solution:

(i) implies (ii) (5 Marks)

(ii) implies (i) (5 Marks)

NB: Suitable stage marks may be fixed by the examiner for alternate correct answers.

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